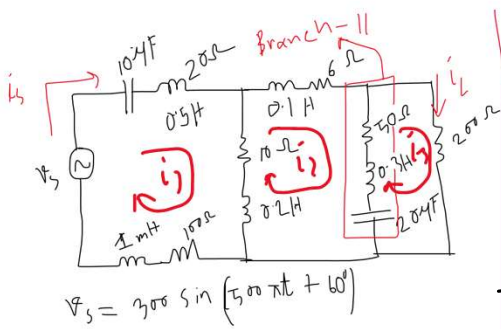




Find →
 (i) $i_s = ?$
 (ii) $i_L = ?$
 (iii) P_s, Q_s, S_s
 (iv) find P, Q, S for Branch-II

$$V_s = 300 \angle 60^\circ$$

$$10 \mu F \rightarrow \frac{1}{500 \times 10^{-6}} = 200 \Omega$$

$$20 \mu F \rightarrow \frac{1}{500 \times 2 \times 10^{-6}} = 100 \Omega$$

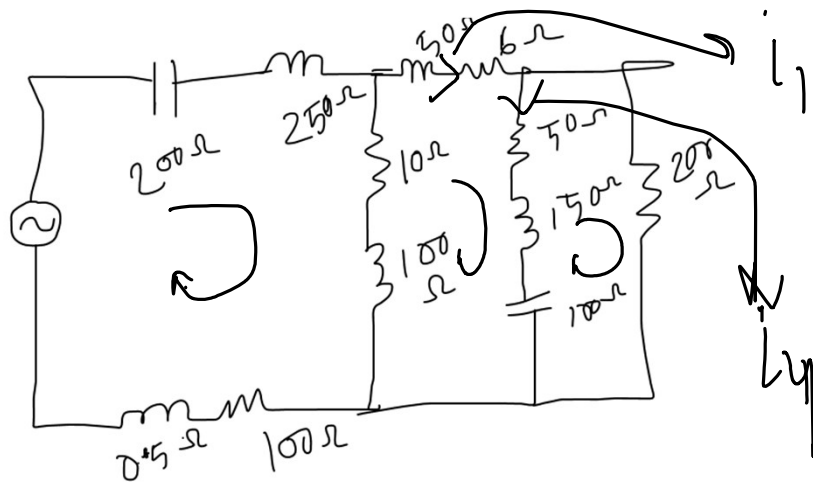
$$0.5 H \rightarrow 0.5 \times 500 = 250 \Omega$$

$$0.2 H \rightarrow 500 \times 0.2 = 100 \Omega$$

$$0.1 H \rightarrow 500 \times 0.1 = 50 \Omega$$

$$0.3 H \rightarrow 500 \times 0.3 = 150 \Omega$$

$$1 mH \rightarrow 500 \times 1 m = 0.5 \Omega$$



$$Z_{Br-2} = 50 + j150 - j100$$

$$= 50 + j50$$

$$I_1 = \frac{(10 + j100)}{(10 + j100) + 50 + j50}$$

$$= 0.52 \angle 13.26^\circ$$

$$I_4 = \frac{200}{200 + 50 + j50} \times I_1$$

$$= 0.41 \angle 1.95^\circ$$

$$V_{Br-2} = I_4 \times Z_{Br-2}$$

$$= 28.84 \angle 47^\circ$$

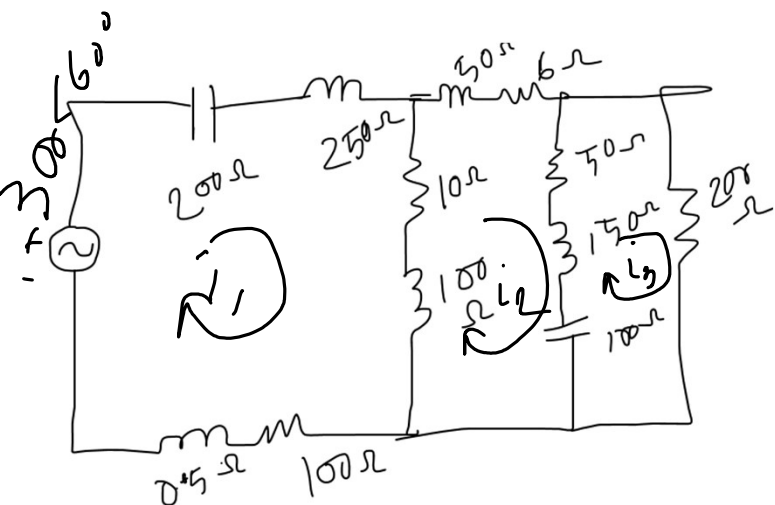
$$\begin{aligned}
 \text{pf}_{Br-2} &= \cos(\theta_{V_{Br-2}} - \theta_{I_4}) \\
 &= \cos(41.7^\circ - 1.95^\circ) \\
 &= 0.71
 \end{aligned}$$

$$\begin{aligned}
 P_{Br-2} &= |V_{Br-2}| |I_{Br-2}| \cos(\theta_{V_{Br-2}} - \theta_{I_4}) \\
 &= 24.84 \times 0.41 \times 0.71
 \end{aligned}$$

$$\overline{S_{Br-2}} = P_{Br-2} + j Q_{Br-2}$$

$$S_{Br-2} = |\overline{S_{Br-2}}|$$

$$\begin{aligned} Q_{Br-2} &= |V_{Br-2}| |I_{Br-2}| \sin(\theta_{V_{Br-2}} - \theta_{I_4}) \\ &= 24.84 \times 0.41 \times \sin(45.05) \end{aligned}$$



Mesh Analysis

Mesh - 1

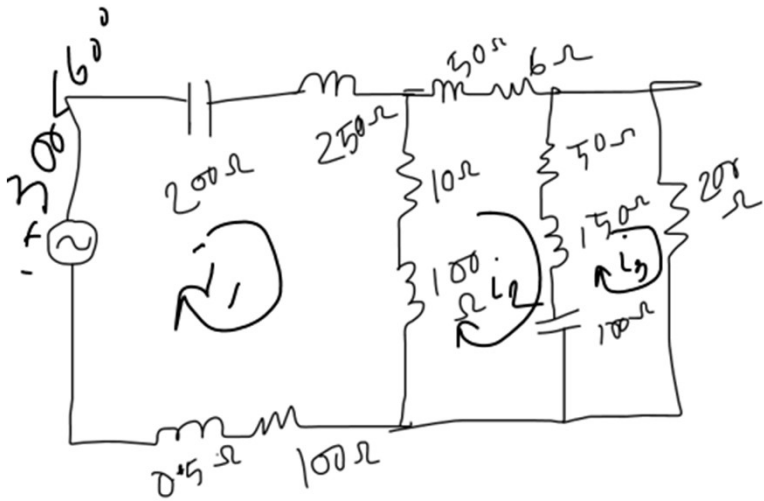
$$\sum V = 0$$

$$-300 \angle 60^\circ + \dot{I}_1(j50) + (\dot{I}_1 - \dot{I}_2)(10 + j100) + \dot{I}_1(100 + j0.5) = 0$$

$$I_1 (110 + j150.5) + I_2 (-10 - j100) = 300 \angle 60^\circ$$

— (1)

Mesh - 2

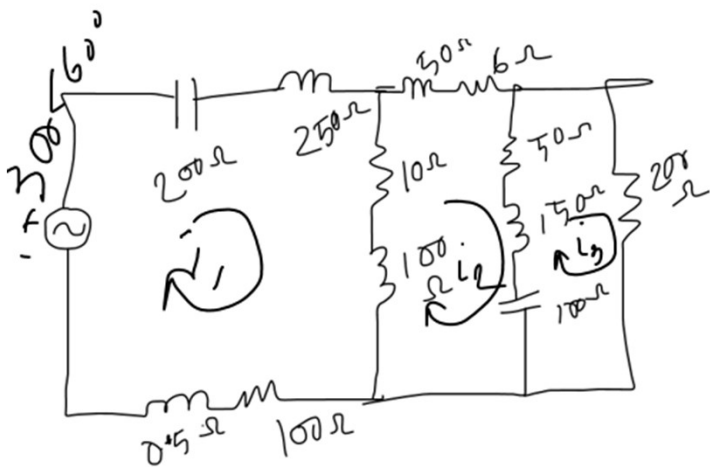


$$(I_2 - I_1)(10 + j100) + (6 + j50)I_2 + (I_2 - I_3)(50 + j50) = 0$$

$$\Rightarrow I_1(-10 - j100) + I_2(60 + j200) + I_3(-50 - j50)$$

$$= 0$$

_____ (2)



Mesh-3

$$(I_3 - I_2)(50 + j50) +$$

$$200 \times I_3 = 0$$

$$\Rightarrow I_2(-50 - j50) + I_3(250 + j250)$$

$$= 0$$

3

Solving eqⁿ (1), (2), (3),

} VNE Crammer's
Rule.

$$I_1 =$$

$$I_2 =$$

$$I_3 =$$

Super-Mesh



When there is a current source between 2 meshes



2 meshes are treated as 1 super mesh

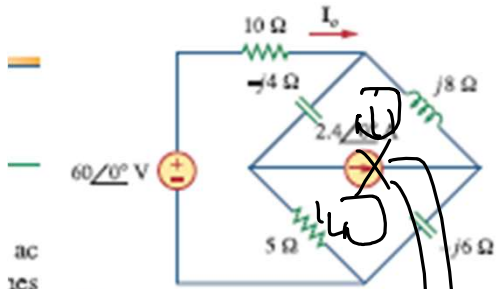


Figure 10.11
For Practice Prob. 10.4.

$$i_2 - i_1 \rightarrow$$

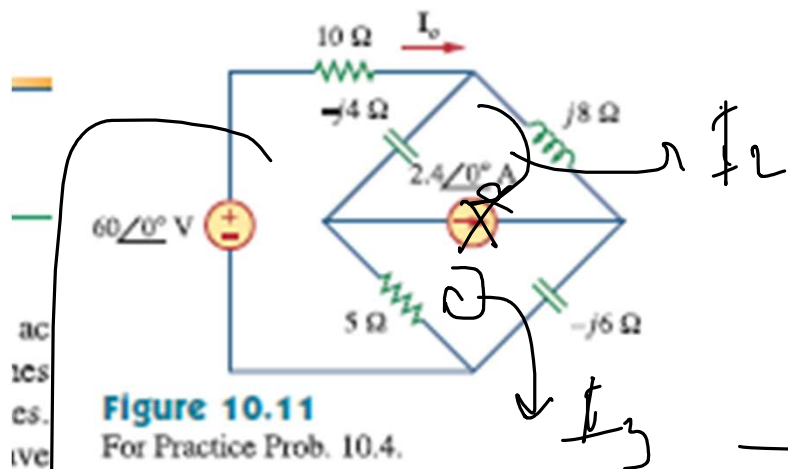


Figure 10.11
For Practice Prob. 10.4.

$$I_3 - I_2 = 2.4 \angle 0^\circ \text{ A} \quad \text{--- ①}$$

Mesh-1

$$-60 \angle 0^\circ + 10 I_1 + (I_1 - I_2)(-j4) + I_1 \times (I_1 - I_3) = 0$$

$$\text{--- ②}$$

9 KVL at supermesh

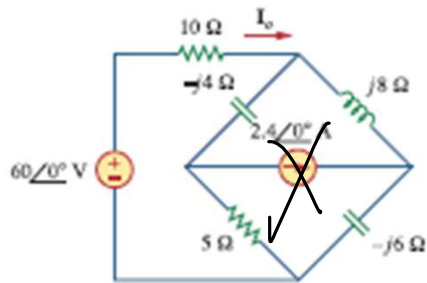


Figure 10.11
For Practice Prob. 10.4.

$$(I_2 - I_1)(-j4) + I_2 \times j8$$

$$+ I_3(-j6) + (I_3 - I_1) \times 5 = 0$$

— (3)

Solving eqⁿ ①, ②, ③.

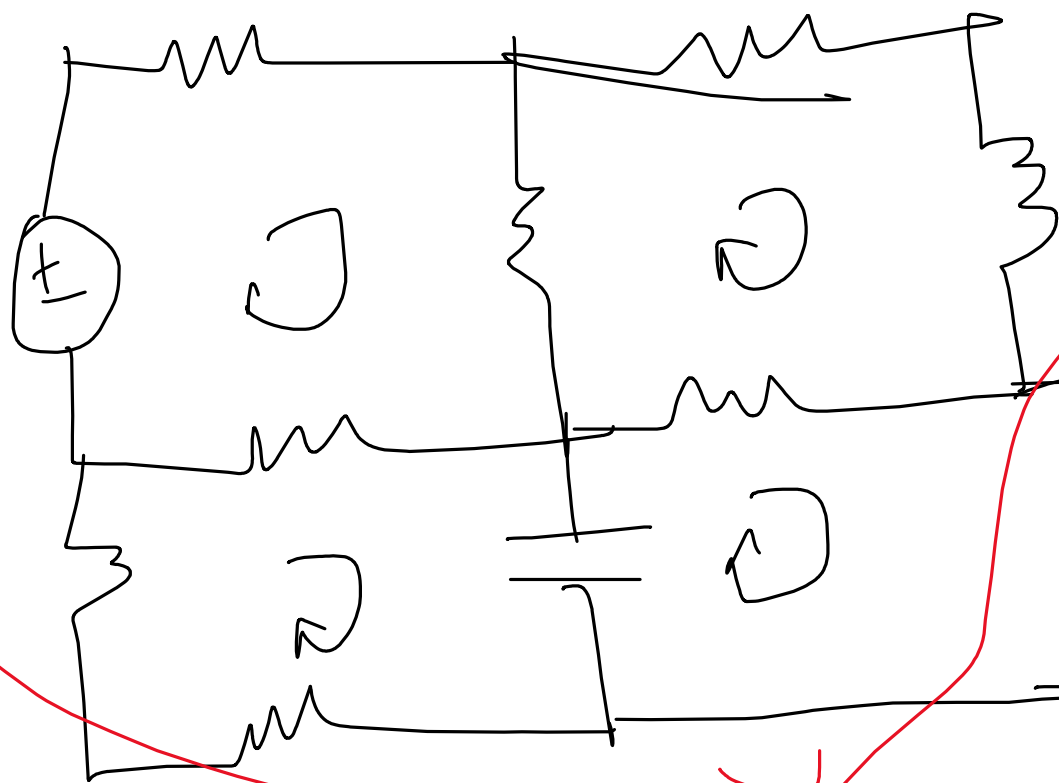
Use Grammar's
Rule.

$$I_1 =$$

$$I_2 =$$

$$I_3 =$$

The unity Theorem



V_{km} z_{km}

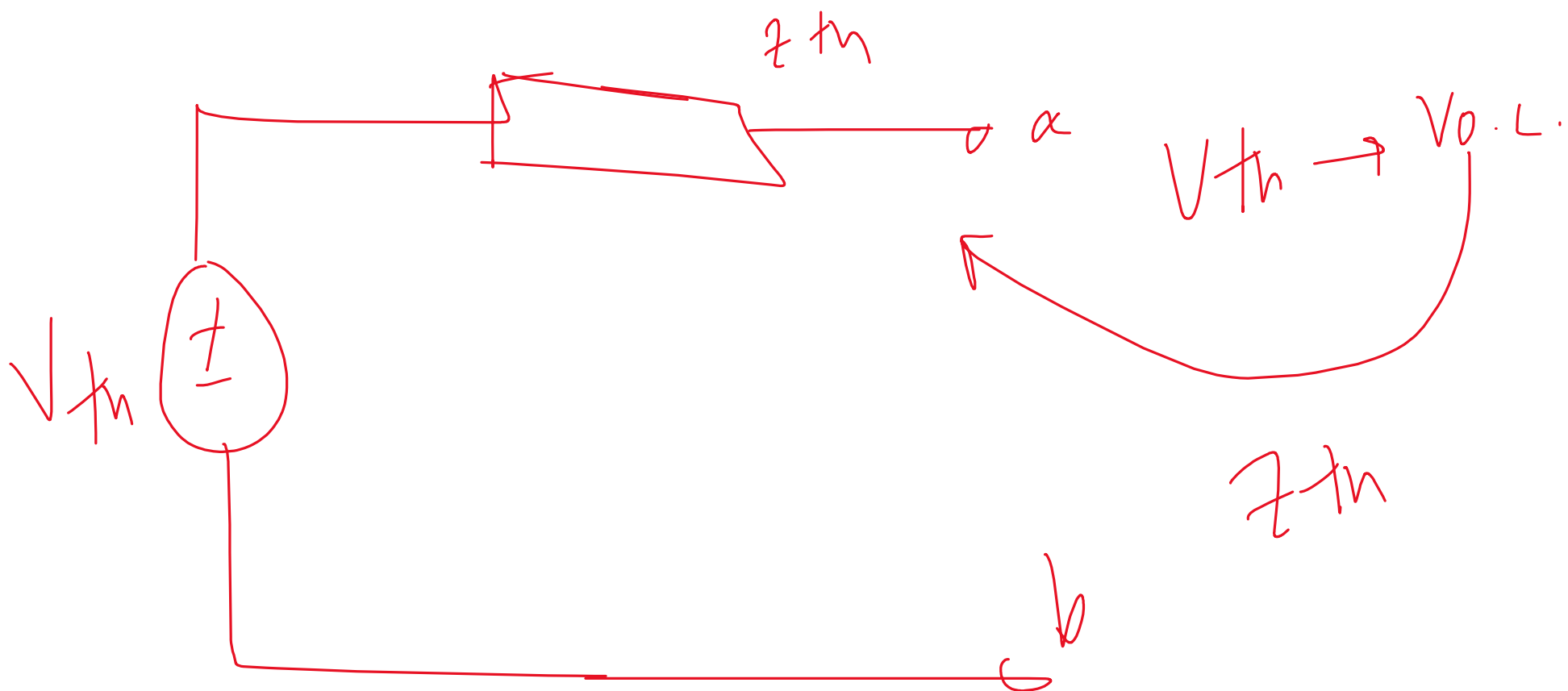
$$V_{OC} = V_{th}$$

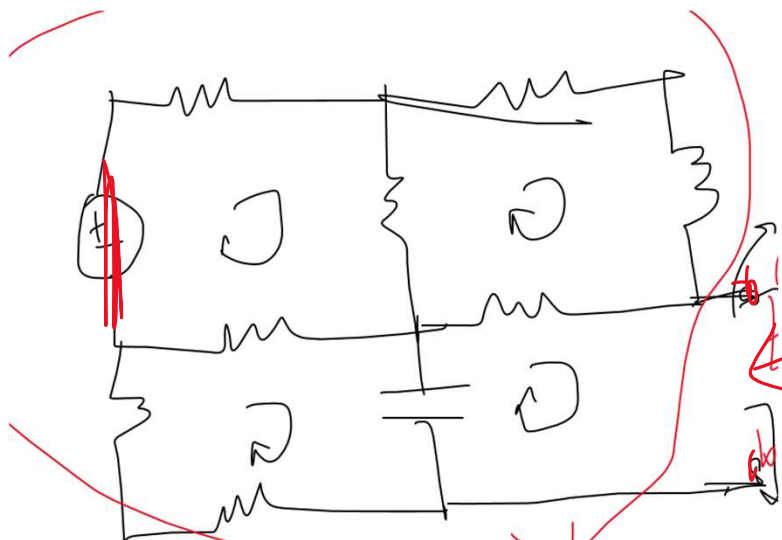
5105

$\Gamma_{1-\alpha} - 1 \rightarrow 0$

20 52

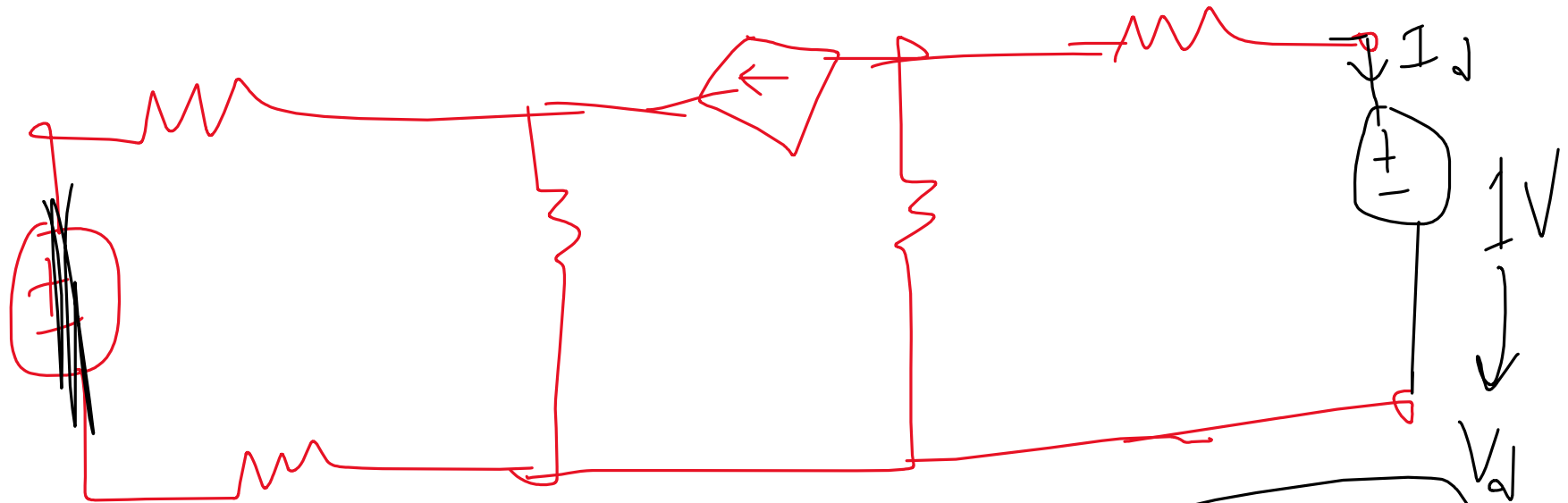
30~





$$Z_m = Z_{o.c.} = Z_{eq}$$

Voltage source $\xrightarrow{\text{turn off}}$ s.c.
 current " $\rightarrow$ o.c.



$V_{th} = V_{oc}$

Z_{th}

$$Z_{th} = \frac{I_d}{V_d}$$

10.67 Find the Thevenin and ~~Norton~~ equivalent circuits at terminals $a-b$ in the circuit of Fig. 10.110.

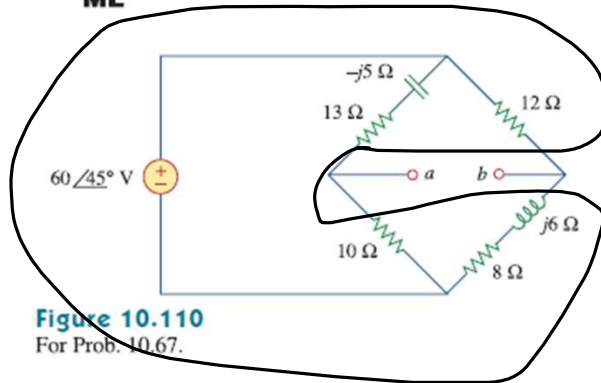
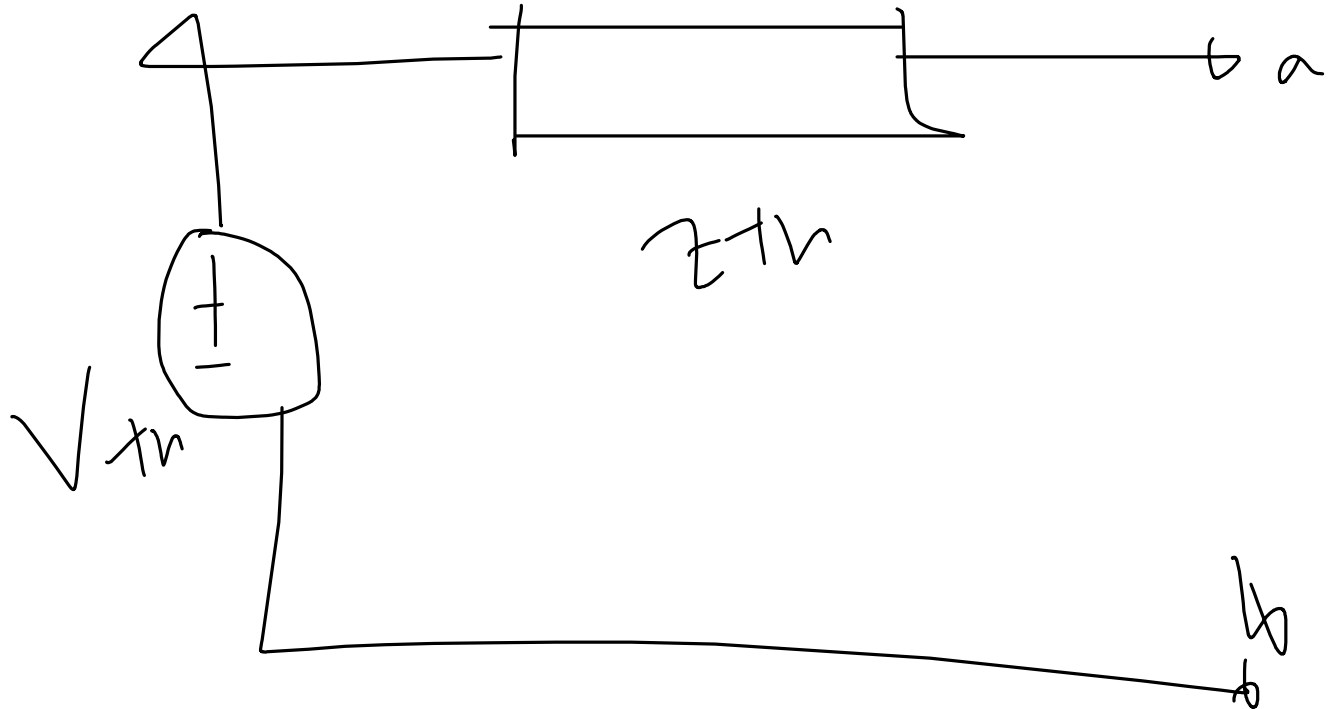


Figure 10.110
For Prob. 10.67.



10.67 Find the Thevenin and Norton equivalent circuits at terminals $a-b$ in the circuit of Fig. 10.110.

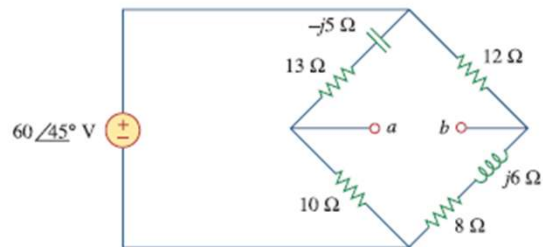


Figure 10.110
For Prob. 10.67.

10.67 Find the Thevenin and Norton equivalent circuits at terminals $a-b$ in the circuit of Fig. 10.110.

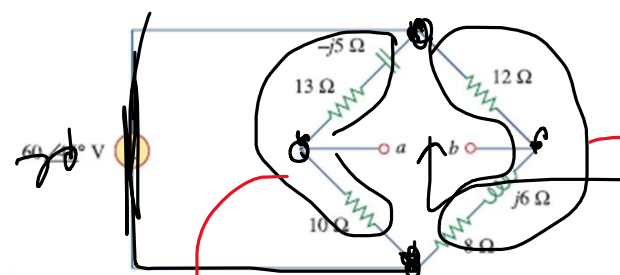


Figure 10.110
For Prob. 10.67.

Z_{th}

$$Z_{eq} = Z_{th}$$

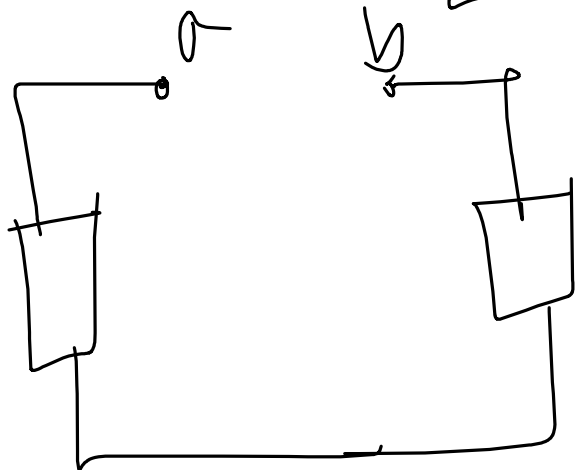
Z_{th}

$$Z_{eq} =$$

$$(13 - j5) \parallel 10$$

$$12 \parallel (8 + j6)$$

$$= 11.3 \angle 5.5^\circ$$



10.67 Find the Thevenin and Norton equivalent circuits at terminals $a-b$ in the circuit of Fig. 10.110.
ML

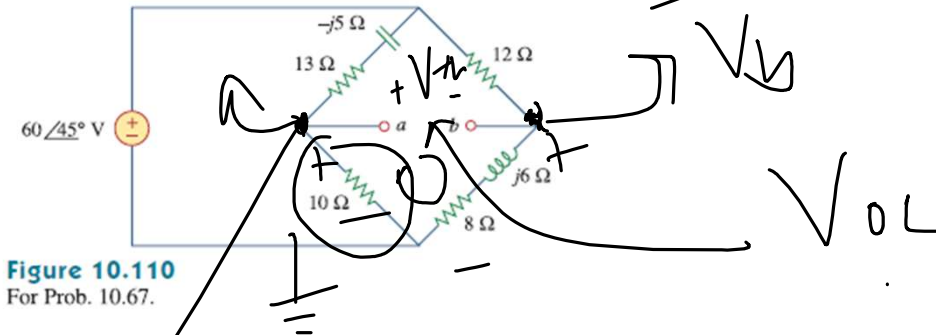


Figure 10.110
For Prob. 10.67.

$$-V_a + V_{Th} + V_b = 0$$

$$\therefore V_{Th} = V_a - V_b \quad \checkmark$$

$$V_{Th} = V_{OL}$$

$$V_a =$$

$$V_a = V_{10\Omega} = \frac{10}{10 + (13 - j5)} \times 60 \angle 45^\circ$$

$$V_b = V_{(8+j6)} = \frac{8 + j6}{8 + j6 + 12} \times 60 \angle 45^\circ$$